SURJECTIONS ONTO COHERENT SHEAVES FROM BOX PRODUCTS

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Allowed standard prerequisites (editorial).

- Homs over opens. coherent-lemma-homs-over-open (cross-chapter, named). Extends sections from U x V via powers of the ideal.
- Finite generation. properties-lemma-finite-type-module (cross-chapter, named). Finite many sections cover each affine patch.

Source reference. Original reference 01YB: focused-seeds/geometry/S9-eqv-on-product/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01PB: focused-seeds/geometry/S9-eqv-on-product/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let R be a Noetherian ring. Let X, Y be finite type schemes over R having the resolution property. For any coherent $\mathcal{O}_{X\times_R Y}$-module $\mathcal{F}$ there exist a surjection $\mathcal{E}\boxtimes\mathcal{G}\to\mathcal{F}$ where $\mathcal{E}$ is a finite locally free $\mathcal{O}_X$-module and $\mathcal{G}$ is a finite locally free $\mathcal{O}_Y$-module.*

Proof. Let $U\subset X$ and $V\subset Y$ be affine open subschemes. Let $\mathcal{I}\subset\mathcal{O}_X$ be the ideal sheaf of the reduced induced closed subscheme structure on $X\setminus U$. Similarly, let $\mathcal{I}'\subset\mathcal{O}_Y$ be the ideal sheaf of the reduced induced closed subscheme structure on $Y\setminus V$. Then the ideal sheaf

$$\mathcal{J}=\mathrm{Im}(\mathrm{pr}_1^*\mathcal{I}\otimes_{\mathcal{O}_{X\times_R Y}}\mathrm{pr}_2^*\mathcal{I}'\to\mathcal{O}_{X\times_R Y})$$

satisfies $V(\mathcal{J})=X\times_R Y\setminus U\times_R V$. For any section $s\in\mathcal{F}(U\times_R V)$ we can find an integer $n>0$ and a map $\mathcal{J}^n\to\mathcal{F}$ whose restriction to $U\times_R V$ gives s, see Cohomology of Schemes, Lemma 01YB. By assumption we can choose surjections $\mathcal{E}\to\mathcal{I}$ and $\mathcal{G}\to\mathcal{I}'$. These produce corresponding surjections

$$\mathcal{E}\boxtimes\mathcal{G}\to\mathcal{J}\quad\text{and}\quad\mathcal{E}^{\otimes n}\boxtimes\mathcal{G}^{\otimes n}\to\mathcal{J}^n$$

and hence a map $\mathcal{E}^{\otimes n}\boxtimes\mathcal{G}^{\otimes n}\to\mathcal{F}$ whose image contains the section s over $U\times_R V$. Since we can cover $X\times_R Y$ by a finite number of affine opens of the form $U\times_R V$

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and since $\mathcal{F}|_{U \times_R V}$ is generated by finitely many sections (Properties, Lemma 01PB) we conclude that there exists a surjection

$$\bigoplus\nolimits_{j=1,\ldots,N} \mathcal{E}_j^{\otimes n_j} \boxtimes \mathcal{G}_j^{\otimes n_j} \to \mathcal{F}$$

where $\mathcal{E}_j$ is finite locally free on X and $\mathcal{G}_j$ is finite locally free on Y. Setting $\mathcal{E} = \bigoplus \mathcal{E}_j^{\otimes n_j}$ and $\mathcal{G} = \bigoplus \mathcal{G}_j^{\otimes n_j}$ we conclude that the lemma is true. $\square$